Python Basics

A beginner-friendly guide to core Python programming concepts.

1. What is Python?

Python is a high-level, general-purpose programming language with readable syntax.

Python is used in web development, automation, data analysis, artificial intelligence, and machine learning.

2. Variables and Data Types

A variable is a name that refers to a value. Example: age = 21.

Common data types include int for whole numbers, float for decimals, str for text, and bool for True or False.

3. Conditional Statements

Conditional statements let a program make decisions. Python uses if, elif, and else.

Example: if marks >= 40: print('Pass') else: print('Fail').

4. Loops

Loops repeat code. A for loop iterates over a sequence; a while loop repeats while its condition is true.

Example: for number in range(3): print(number).

5. Functions

A function is a reusable block of code defined with def. Example: def add(a, b): return a + b.

Functions organize code and reduce repetition.

6. Lists and Dictionaries

A list stores an ordered collection, such as numbers = [10, 20, 30].

A dictionary stores key-value pairs, such as student = {'name': 'Asha', 'marks': 85}.